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MATHEMATICS
SHORT ANSWER PROBLEMS

Name : _____ **Index Number** : _____

Country : _____



18th International Mathematics and Science Olympiad

Indonesia

15 December 2021

Instructions:

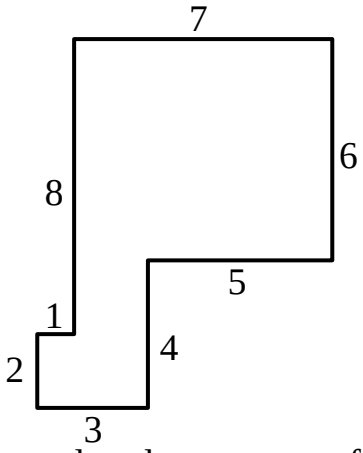
1. Write your name, country and index number on both the Question Booklet and the Answer Sheet.
2. Write your Arabic Numerical answers or English alphabet answers only in the Answer Sheet.
3. There are 25 questions in this paper.
4. For problems involving more than one answer, marks are only awarded when ALL answers are correct.
5. Each question is worth 1 mark. There is no penalty for a wrong answer.
6. You have 60 minutes to complete this paper.
7. Use black pen or blue pen or pencil to write your answer.

Do not turn over this page until you are told to do so.

SHORT ANSWER PROBLEMS

1. At a New Year's Eve party, each member of the Math club shakes hands with exactly 20 members of the IT club, while each member of the IT club shakes hands with exactly 15 members of the Math club. If the two clubs have a total of 84 members that attend the party, how many more IT club members than Math club members are there in the party?

2. A carpenter builds an octagonal board, having side lengths of 1, 2, 3, 4, 5, 6, 7 and 8 cm respectively, as shown on the diagram. If the angles of the polygon are all right, then what is the area, in cm^2, of the board?

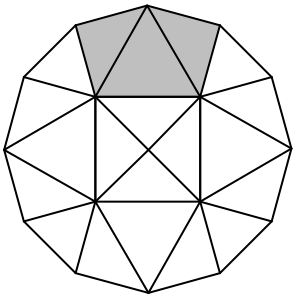


3. Five cubes with edges 1 cm, 2 cm, 3 cm, 4 cm and 5 cm are placed one on top of another in such a way that the largest cube is located at the bottom most, while the smallest cube is located at the topmost, which forms a tower. What is the total visible surface area, in cm^2, of the tower, excluding the bottom face?

4. A rabbit and a turtle competed in a 10000 m race, where the rabbit's speed is five times the turtle's speed. From the start, the turtle was moving at a constant speed, without rest all throughout, while the rabbit, knowing that he was in the lead, stopped and slept for some time. Upon waking up and knowing that the turtle was already 5000 m ahead of him, the rabbit started to run again. However, the rabbit was still 100 m away, when the turtle reached the finish line. How long did the turtle travel, in m, while the rabbit was asleep?

5. A wolf and a dog are rivals, where they bark at each other every time they meet. One day, they saw each other from far away and the total distance between them would take the wolf to run a total of 568 steps. It is known that the time for a wolf to run 9 steps is equal to the time for a dog to run 7 steps and the distance for a wolf to run 5 steps is equal to the distance for a dog to run 4 steps. How many steps will the dog have to run towards the direction of the wolf for them to meet?

6. In the diagram on the right, a regular dodecagon (12-gon) is divided into some number of right isosceles and equilateral triangles. What is the ratio between its shaded area and unshaded area?



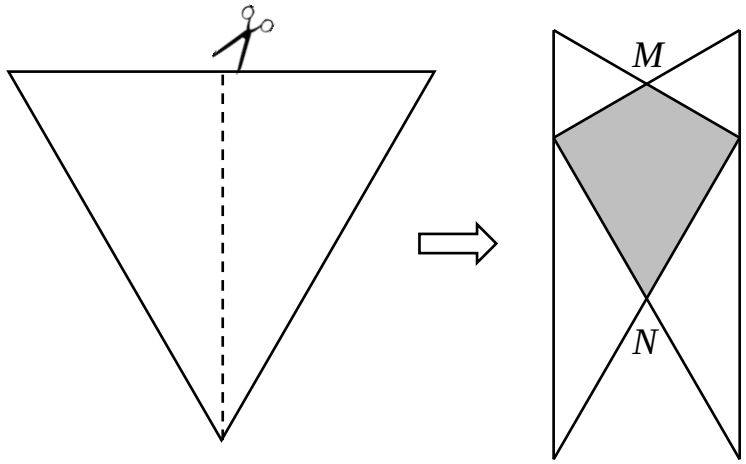
7. There are n lines found on a plane that intersect exactly 7 other lines. Find the sum of all possible values of n.

8. Let a, b, c and d be four different prime numbers that satisfy

$$500a+101b+101c+d=2021.$$

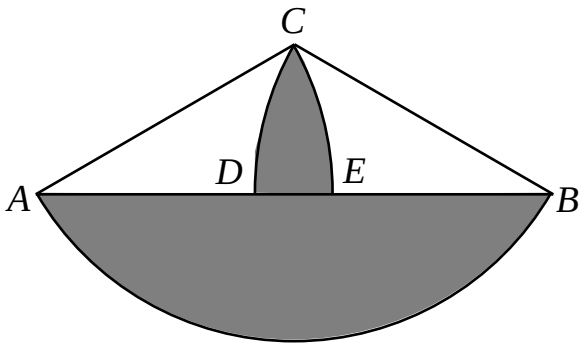
What is the value of $a+b+c+d$?

9. An equilateral triangle is cut into two smaller triangles as shown in the diagram. Now, these two shapes are overlapped to form a new diagram such that intersection points M and N are the midpoints of the sides of the triangles. What is the ratio between its shaded area and unshaded area of the new diagram?



10. There are five bags, each containing 15 coins. One bag contains only gold coins, another one contains only silver coins, the third one contains only bronze coins, and each of the remaining two contains 5 gold, 5 silver and 5 bronze coins. What is the minimum number of coins that must be drawn, from any combination of bags, in order to determine the content of at least one bag?

11. In the diagram on the right, ABC is an isosceles triangle with $AC=BC=21\text{cm}$ and $\angle ACB=120°$. What is the positive difference of the areas, in cm^2, of the two shaded regions? (Use $\pi=\frac{22}{7}$)



12. A written test has 30 multiple-choice questions. A student receives 5 points for each correct answer, 1 point for each question left blank and 0 points for each incorrect answer. Suppose that at the end of the written test, five students made the following statements:

Alex says: "My test score is 147."
Bob says: "My test score is 144."
Chris says: "My test score is 143."
Daniel says: "My test score is 141."
Eric says: "My test score is 139."

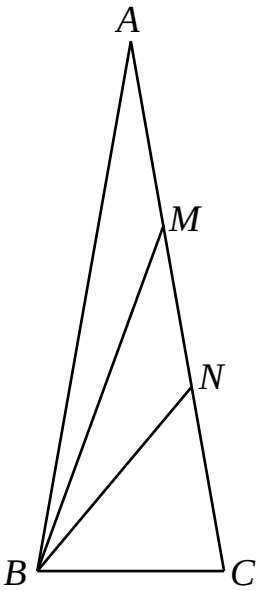
Only one of these students could be correct. Who is he?

13. Five different positive integers are to be arranged in a sequence with the following conditions:

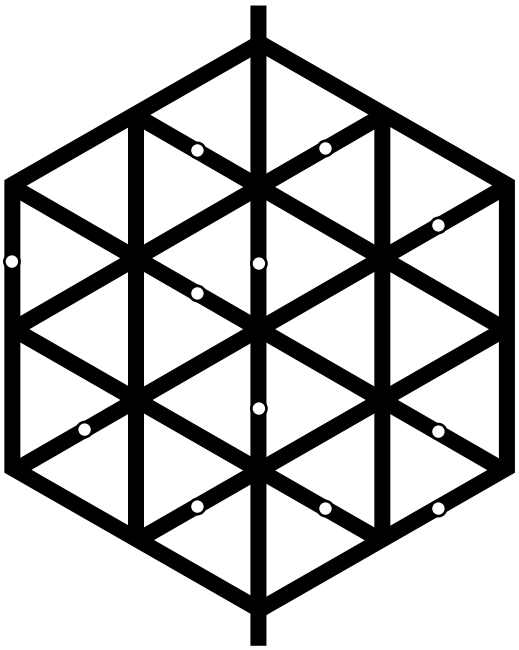
- There are 3 even numbers.
- There are 2 prime numbers.
- The sum of the 1st and 2nd numbers is odd, the sum of the last and 2nd to the last numbers is also odd and the sum of the three middle numbers is 10.
- The 1st and 2nd numbers are known to be prime, while the 3rd number is 6.

What is the 2nd number of the sequence?

14. In the diagram below, ABC is an isosceles triangle, where $\angle A = 20°$. Let M and N be two points on AC such that $AM = CN = CB$. What is the measure, in degrees, of $\angle MBN$?



15. The diagram below shows a hexagonal garden with several paths. There are twelve apples, each marked by white circles, lying along some of the paths. You may enter the garden from either end and exit on the other end, and move along the paths to collect as many apples as you can. However, you are not allowed to pass through any path or intersections more than once. What is the maximum number of apples you can collect?



16. There are 60 coins on a table. Eric and Fred play a game where they take turns removing coins from the table. The number of coins removed on each turn must be between 1 and 8, inclusive. The player who makes the last move wins. If Eric starts first, how many coins does he need to remove from the table on his first turn to be able to guarantee his win?

17. The sixth grade classes of a certain school are to be re-organized from class A and B into classes X, Y and Z by transferring $\frac{1}{3}$ of the students from class A and $\frac{1}{4}$ of the students from class B to form class X, then followed by transferring $\frac{1}{4}$ of the students from class A and $\frac{1}{3}$ of the students from class B to form class Y, then the remaining 30 students of class A and B will form class Z. If the number of students in class X is $\frac{1}{10}$ more than the number of students in class Y, how many students are there in class A?

18. Let $M = (1+2+3+4+5)+(1^2+2^2+3^2+4^2+5^2)+...+(1^{10}+2^{10}+3^{10}+4^{10}+5^{10})$. What is the units digit of *M*?

19. As shown in the diagram below, each cell of an 11×11 board is either filled-up with a letter or a number from "IMSO18" or is coloured black. An ant starts from letter "I" which is located at the center of the board. In each step, it can choose to move vertically, horizontally or diagonally to any neighboring cell. It will only stop moving when it reaches a boundary cell. Find the number of routes such that the letters in cells along the route will read exactly as "IMSO18" and no black cell is on the route.

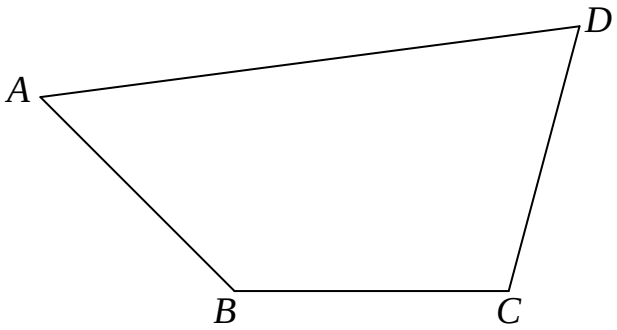
8	8	8	8	8	8	8	8	8	8	8
8	■	1	1	1	1	1	1	1	■	8
8	1	O	■	O	O	O	■	O	1	8
8	1	■	S	S	■	S	S	■	1	8
8	1	O	S	M	M	M	S	O	1	8
8	1	O	■	M	I	M	■	O	1	8
8	1	O	S	M	M	M	S	O	1	8
8	1	■	S	S	■	S	S	■	1	8
8	1	O	■	O	O	O	■	O	1	8
8	■	1	1	1	1	1	1	1	■	8
8	8	8	8	8	8	8	8	8	8	8

20. The circumference of a circle is divided into 2021 equal parts by 2021 points. We select and label 4 points randomly with "*I*", "*M*", "*S*" and "*O*".
What is the ratio between the number of selections with intersecting segments *IM* and *SO* to the number of selections with non-intersecting segments *IM* and *SO*?

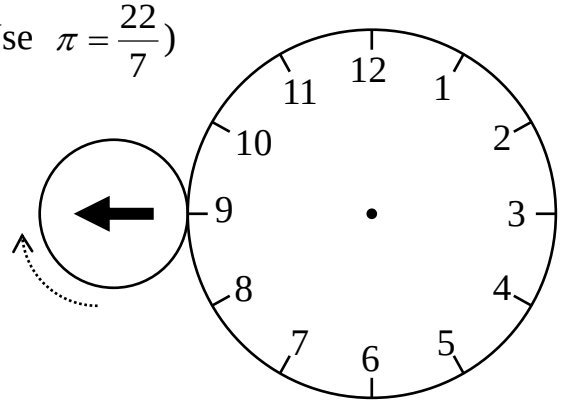
21. It is given that the sum $n+2n+3n+...+9n$ produces a number whose digits are all the same, where *n* is the smallest positive integer satisfying the above condition. What is the sum of digits of *n*?

22. There is a 3-digit number $N=\overline{ZXY}$, where $X>Y>Z>0$. Let *P* be the largest possible 3-digit number formed using the digits of *N* and *Q* be the smallest possible 3-digit number formed using the digits of *N*. If $P-Q=N$, then find the value of $X+Y+Z$.

23. In the diagram below, $AB=BC=CD$, $\angle ABC=135°$ and $\angle BCD=105°$.
What is the measure, in degrees, of $\angle ADC$?



24. A circular disc rolls around a fixed circular clock in a clockwise direction. The radius of the disc and the clock are 21 cm and 42 cm, respectively. In the initial position, the disc is at the 9-hour mark and the arrow on it points to the left, as shown in the diagram below. What is the shortest distance, in cm, that the disc must roll on the circumference of the clock so that the arrow points to the same direction again? (Use $\pi=\frac{22}{7}$)



25. Angel and Petko plays a game using two standard dice. In each turn, the players take turns in rolling the two dice at the same time, where the two resulting numbers are added together to get the score. Angel will be the winner if she gets a higher score than Petko. Find the ratio between the cases when Angel wins and the cases when Angel does not win.

18th International Mathematics and Science Olympiad

SHORT ANSWER PROBLEMS

ANSWER SHEET

Name : _____ **Index Number** : _____

Country : _____ **Score** : _____

1.	members	***14.***	°
2.	cm^2	***15.***	apples
3.	cm^2	***16.***	coins
4.	m	***17.***	students
5.	steps	***18.***	
6.	:	***19.***	routes
7.		***20.***	:
8.		***21.***	
9.	:	***22.***	
10.	coins	***23.***	°
11.	cm^2	***24.***	cm
12.		***25.***	:
13.			